

EXP 4

Objective: 1

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('--- Rectangular Window Coefficients ---');

disp(b_rect);

%% --- 2. FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm);

%% --- 3. FIR Low Pass using Hanning Window ---

b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');

disp(b_hann);

%% --- Plot Frequency Responses ---

figure;

freqz(b_rect,1);

title('Low-pass FIR using Rectangular Window');

figure;

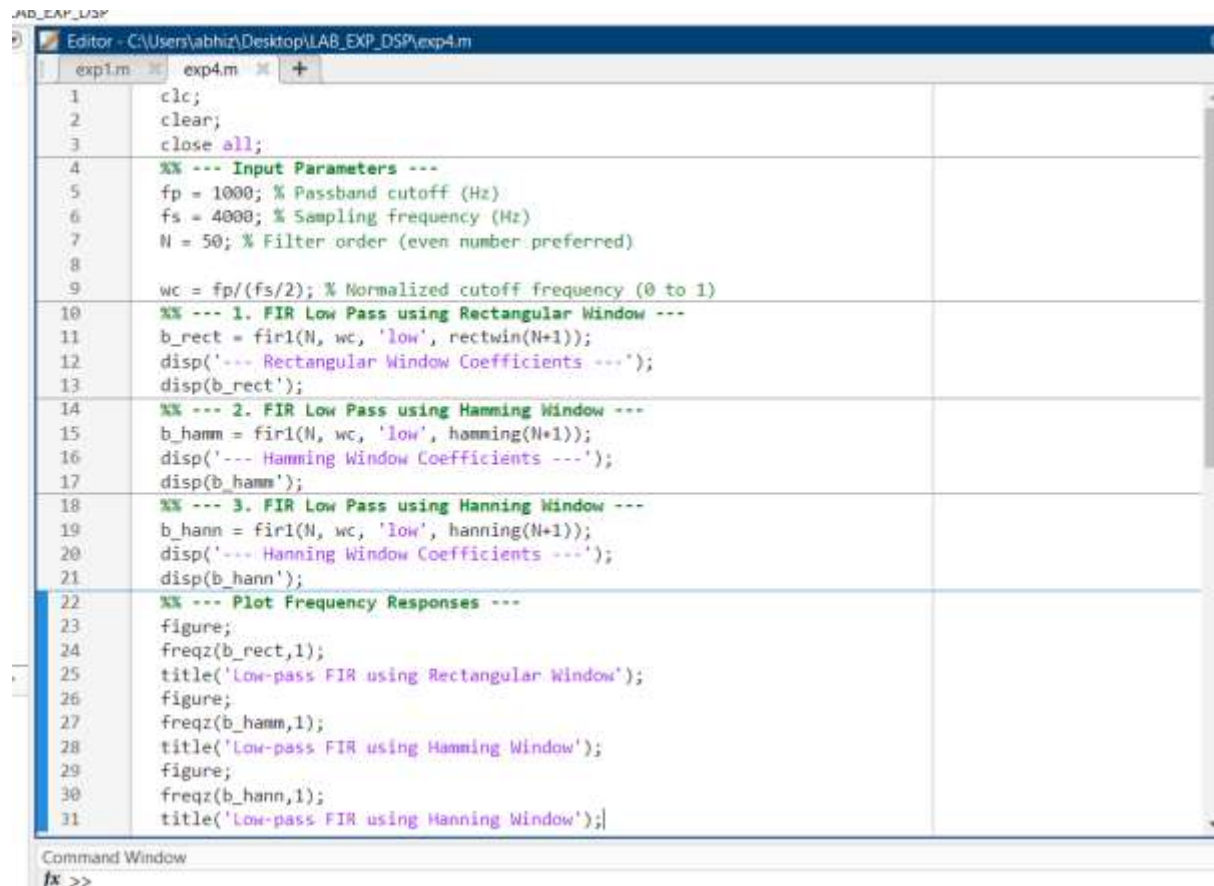
freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window');

figure;
```

```
freqz(b_hann,1);
```

```
title('Low-pass FIR using Hanning Window');
```



```
1  clc;
2  clear;
3  close all;
4  %% --- Input Parameters ---
5  fp = 1000; % Passband cutoff (Hz)
6  fs = 4000; % Sampling frequency (Hz)
7  N = 50; % Filter order (even number preferred)
8
9  wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
10 %% --- 1. FIR Low Pass using Rectangular Window ---
11 b_rect = fir1(N, wc, 'low', rectwin(N+1));
12 disp('--- Rectangular Window Coefficients ---');
13 disp(b_rect);
14 %% --- 2. FIR Low Pass using Hamming Window ---
15 b_hamm = fir1(N, wc, 'low', hamming(N+1));
16 disp('--- Hamming Window Coefficients ---');
17 disp(b_hamm);
18 %% --- 3. FIR Low Pass using Hanning Window ---
19 b_hann = fir1(N, wc, 'low', hanning(N+1));
20 disp('--- Hanning Window Coefficients ---');
21 disp(b_hann);
22 %% --- Plot Frequency Responses ---
23 figure;
24 freqz(b_rect,1);
25 title('Low-pass FIR using Rectangular Window');
26 figure;
27 freqz(b_hamm,1);
28 title('Low-pass FIR using Hamming Window');
29 figure;
30 freqz(b_hann,1);
31 title('Low-pass FIR using Hanning Window');
```

Output

--- Rectangular Window Coefficients ---

0.0126

-0.0000

-0.0137

0.0000

0.0150

-0.0000

-0.0166

0.0000

0.0185

-0.0000

-0.0210

0.0000

0.0242
-0.0000
-0.0286
0.0000
0.0349
-0.0000
-0.0449
0.0000
0.0629
-0.0000
-0.1048
0.0000
0.3145
0.4940
0.3145
0.0000
-0.1048
-0.0000
0.0629
0.0000
-0.0449
-0.0000
0.0349
0.0000
-0.0286
-0.0000
0.0242
0.0000
-0.0210
-0.0000
0.0185

0.0000
-0.0166
-0.0000
0.0150
0.0000
-0.0137
-0.0000
0.0126

--- Hamming Window Coefficients ---

0.0010
-0.0000
-0.0013
0.0000
0.0021
-0.0000
-0.0034
0.0000
0.0055
-0.0000
-0.0084
0.0000
0.0125
-0.0000
-0.0181
0.0000
0.0260
-0.0000
-0.0379
0.0000
0.0580

-0.0000

-0.1026

0.0000

0.3168

0.4995

0.3168

0.0000

-0.1026

-0.0000

0.0580

0.0000

-0.0379

-0.0000

0.0260

0.0000

-0.0181

-0.0000

0.0125

0.0000

-0.0084

-0.0000

0.0055

0.0000

-0.0034

-0.0000

0.0021

0.0000

-0.0013

-0.0000

0.0010

--- Hanning Window Coefficients ---

0.0000

-0.0000

-0.0004

0.0000

0.0013

-0.0000

-0.0028

0.0000

0.0050

-0.0000

-0.0081

0.0000

0.0122

-0.0000

-0.0179

0.0000

0.0259

-0.0000

-0.0378

0.0000

0.0580

-0.0000

-0.1027

0.0000

0.3172

0.5000

0.3172

0.0000

-0.1027

-0.0000

0.0580

0.0000

-0.0378

-0.0000

0.0259

0.0000

-0.0179

-0.0000

0.0122

0.0000

-0.0081

-0.0000

0.0050

0.0000

-0.0028

-0.0000

0.0013

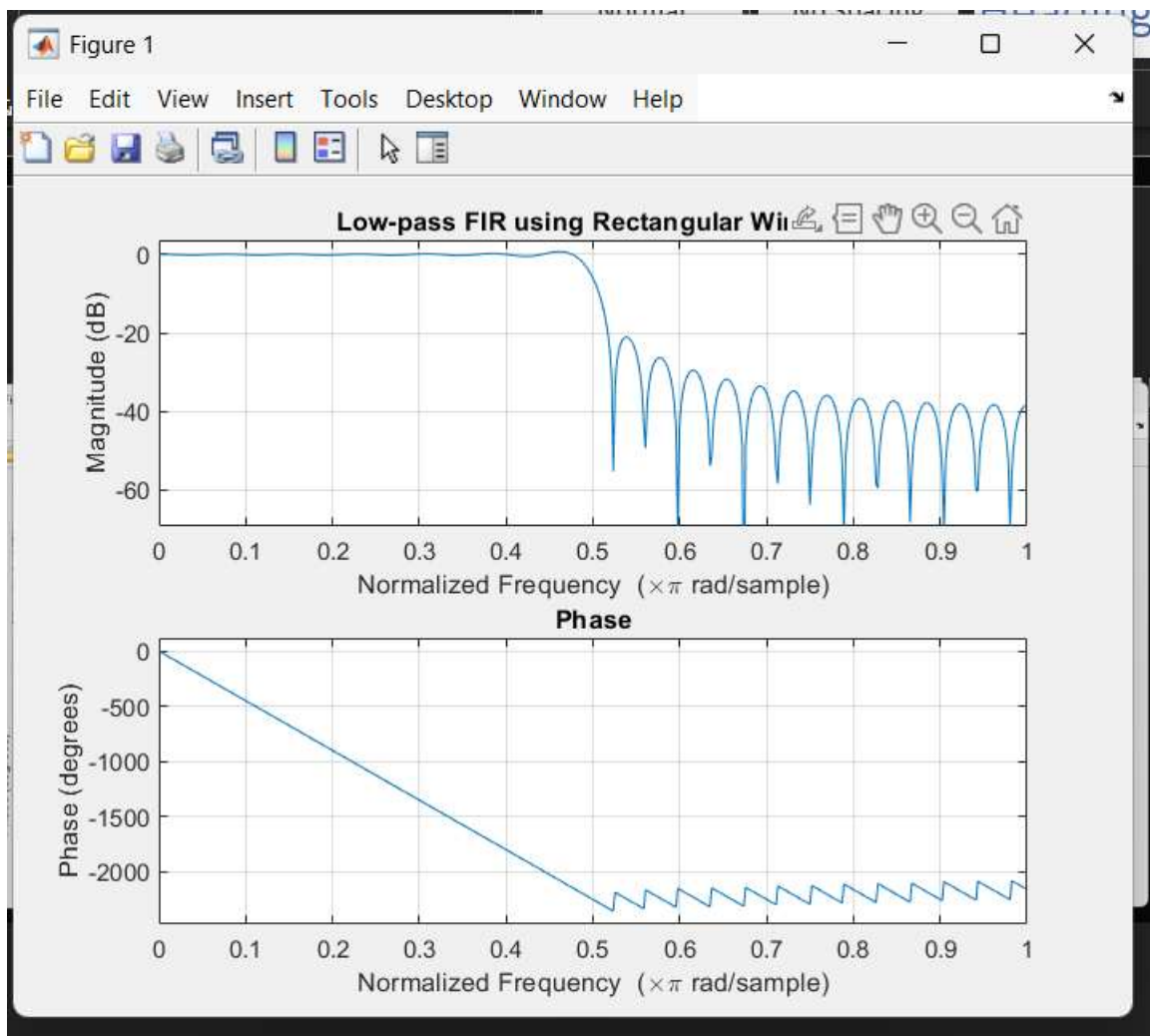
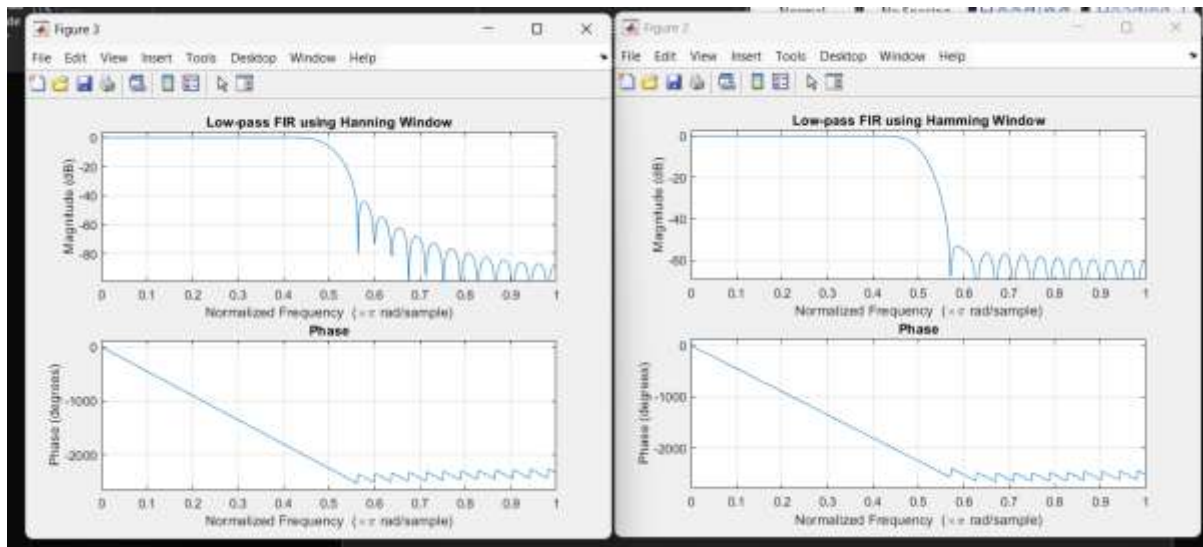
0.0000

-0.0004

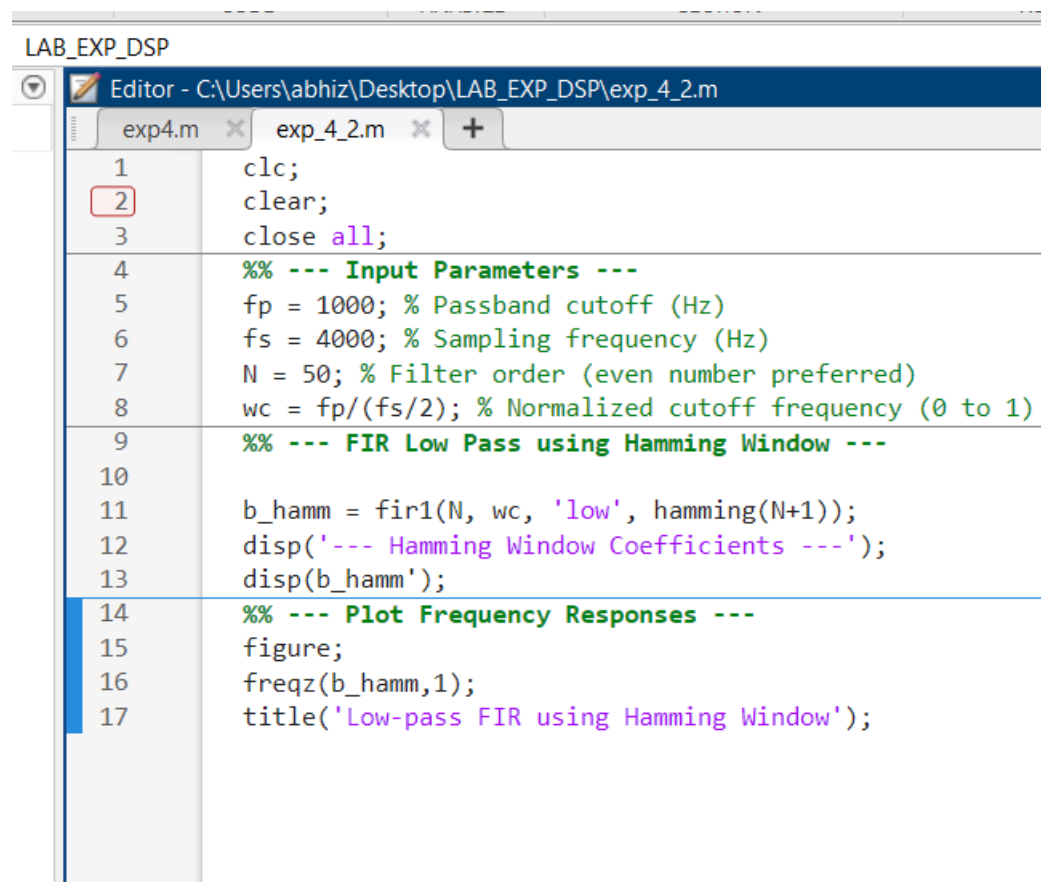
-0.0000

0.0000

>>



Exp 4.2



```
LAB_EXP_DSP
Editor - C:\Users\abhiz\Desktop\LAB_EXP_DSP\exp_4_2.m
exp4.m  exp_4_2.m  +
1      clc;
2      clear;
3      close all;
4      %% --- Input Parameters ---
5      fp = 1000; % Passband cutoff (Hz)
6      fs = 4000; % Sampling frequency (Hz)
7      N = 50; % Filter order (even number preferred)
8      wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9      %% --- FIR Low Pass using Hamming Window ---
10
11      b_hamm = fir1(N, wc, 'low', hamming(N+1));
12      disp('--- Hamming Window Coefficients ---');
13      disp(b_hamm);
14      %% --- Plot Frequency Responses ---
15      figure;
16      freqz(b_hamm,1);
17      title('Low-pass FIR using Hamming Window');
```

Output

--- Hamming Window Coefficients ---

0.0010
-0.0000
-0.0013
0.0000
0.0021
-0.0000
-0.0034
0.0000
0.0055
-0.0000
-0.0084
0.0000
0.0125

-0.0000

-0.0181

0.0000

0.0260

-0.0000

-0.0379

0.0000

0.0580

-0.0000

-0.1026

0.0000

0.3168

0.4995

0.3168

0.0000

-0.1026

-0.0000

0.0580

0.0000

-0.0379

-0.0000

0.0260

0.0000

-0.0181

-0.0000

0.0125

0.0000

-0.0084

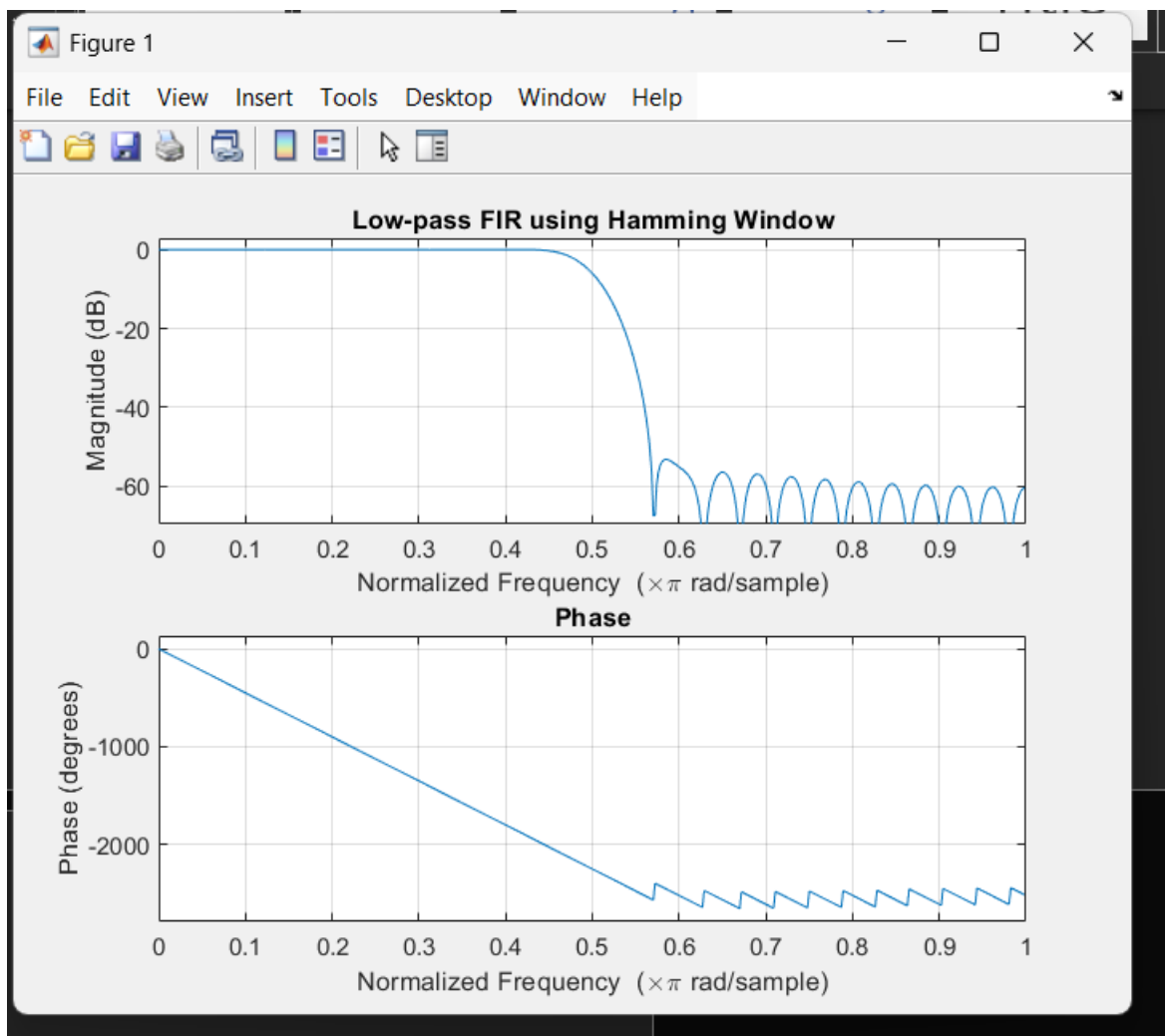
-0.0000

0.0055

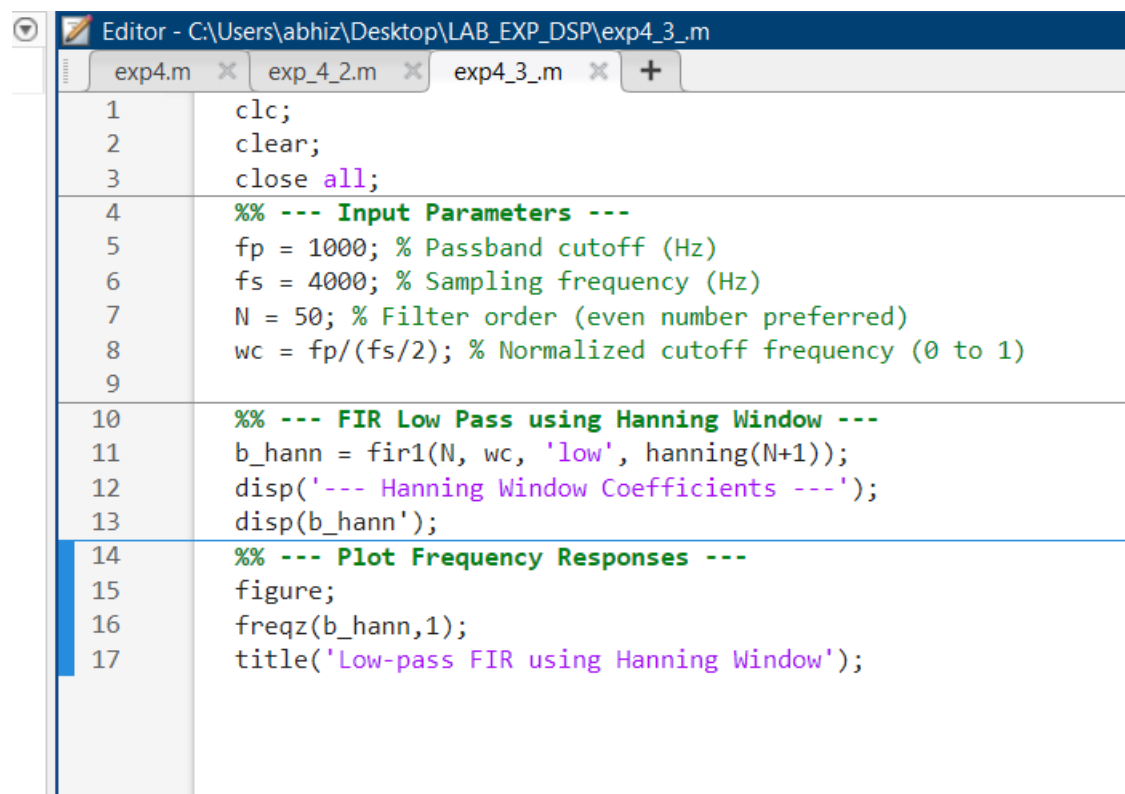
0.0000

-0.0034
-0.0000
0.0021
0.0000
-0.0013
-0.0000
0.0010

>>



Exp 4.3



```
1 clc;
2 clear;
3 close all;
4 %% --- Input Parameters ---
5 fp = 1000; % Passband cutoff (Hz)
6 fs = 4000; % Sampling frequency (Hz)
7 N = 50; % Filter order (even number preferred)
8 wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9
10 %% --- FIR Low Pass using Hanning Window ---
11 b_hann = fir1(N, wc, 'low', hanning(N+1));
12 disp('--- Hanning Window Coefficients ---');
13 disp(b_hann);
14 %% --- Plot Frequency Responses ---
15 figure;
16 freqz(b_hann,1);
17 title('Low-pass FIR using Hanning Window');
```

Output

--- Hanning Window Coefficients ---

0.0000
-0.0000
-0.0004
0.0000
0.0013
-0.0000
-0.0028
0.0000
0.0050
-0.0000
-0.0081
0.0000
0.0122

-0.0000

-0.0179

0.0000

0.0259

-0.0000

-0.0378

0.0000

0.0580

-0.0000

-0.1027

0.0000

0.3172

0.5000

0.3172

0.0000

-0.1027

-0.0000

0.0580

0.0000

-0.0378

-0.0000

0.0259

0.0000

-0.0179

-0.0000

0.0122

0.0000

-0.0081

-0.0000

0.0050

0.0000

-0.0028

-0.0000

0.0013

0.0000

-0.0004

-0.0000

0.0000

>>

